MILITARY SPECIFICATION

PUMP, CIRCULATING, 120/230 Volt, 50/60 Hz AC

This limited coordinated military specification has been prepared by the Defense Personnel Support Center, based on currently available technical information. However, pending its promulgation as a coordinated military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers a circulating impeller type pump for use in a water bath for thawing frozen blood units.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center ATTN: Directorate of Medical Materiel DPSC-RST, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC/NA FSC 6640
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SPECIFICATIONS

FEDERAL

Military

MIL-D-42048 - Deployable Medical Systems, General Requirements for.

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

3. REQUIREMENTS

3.1 General. The complete requirements for acquiring the circulating pump described herein shall consist of this specification and the latest issue of Military Specification MIL-D-42048, except where otherwise indicated.

3.2 Design. Shall be circulating impeller type for use in a temperature controlled water bath for use in thawing frozen blood units. Shall be rated for a delivery capacity of 10 liters (2.6 gallons) per minute at zero head. Shall be electrically operated by thermally protected motor rated for 1550 rpm.

3.3 Construction. Shall be of bronze and brass construction. The motor shall be top mounted, and the bottom shall have a base that encloses the impeller type blades. A vertical shaft, enclosed in a protective housing, shall connect the motor shaft to the impeller type blades. An outlet port for a vertical discharge of water shall be included in the base. The bottom of the base shall have a suitable opening for the inlet water. The base shall be constructed with mounting feet in order that the circulating pump can be positioned in an upright position.

3.4 Electrical characteristics. Unless otherwise indicated in a contract or purchase order, the electrical and line cord characteristics shall conform to requirements contained in the latest edition of MIL-D-42048.

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3.5 Dimensions. Shall have a nominal overall height of 13 inches (34 cm) with at least 9 inches (23 cm) of that height, as measured from the bottom, being submersible in a water bath.

3.6 Workmanship. The circulating pump shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One pump, as specified, constitutes one unit.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 First article inspection. First article inspection shall be performed on the circulating pump when a first article sample is required. This inspection shall include examination and tests.

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4.1.4 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer of supplier, the NSN, the item identification, the name of the component/material, the lot number and the lot size.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I. The unit of product for sampling purposes shall be one circulating pump.

TABLE I. Sampling for examination.

	Inspection level	AQL (percent defects)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted as follows:

a. First article. Where first article sample is required, all tests shall be performed on that unit.

b. Acceptance testing. When a first article sample has not been required, one unit shall be subjected to all tests specified herein. If a first article sample has successfully passed all tests, performance testing shall not be required at acceptance.

4.3 Examination. The circulating pump shall be examined to determine compliance with the requirements of this specification in accordance with the classification of defects in table II.

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Table II. Categories and Defects*

Categories	Defects
Major	
101	Circulating pump not complete with all component parts.
102	Circulating pump not free of wobbling or rocking when placed on a flat horizontal surface.
103	Circulating pump not free of sharp edges or burrs accessible to an operator during normal use or cleaning.
104	Components not usable or properly secured.
105	Power cord not free of cuts, tears, or abrasion.
106	Power cord not properly secured to housing.
107	Blades of power cord not properly aligned.
Minor	
201	Circulating pump not supplied in a clean condition.
202	Circulating pump not free of clusters of chips or scratches.
203	Electric control markings not complete, legible, or permanent.
204	Circulating pump not supplied with required service data that is complete and legible.

*Examination shall not be limited to the classified possible defects listed herein.

4.4 Tests. Tests shall be conducted to determine compliance with this specification.

4.4.1 Operational tests. All circulating pumps shall be tested for compliance with performance requirements. Each circulating pump shall be connected to a suitable power source, in accordance with the manufacturer's instructions, and operated to demonstrate the proper functioning of mechanical and electrical components. Any evidence of or tendency to failure of any component shall be cause for rejection.

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5. PACKAGING

5.1 General.

The complete requirements for packaging shall consist of this specification and the latest issue of Military Specification MIL-D-42048 except where otherwise indicated.

6. NOTES

6.1 Contractor certification. The contractor shall certify that the product offered meets the salient characteristics of this description and conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

6.2 Regulatory requirements.

6.2.1 Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

6.2.2 Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Public Law 94-580, as amended, and as implemented by Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

6.2.3 Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order. Provisioning technical documentation required and appropriate Data Item Descriptions (DID's) should be included as part of the contract.

6.2.4 Applicable NSN. This specification covers the requirements for the following item:

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NSN	Item Identification
6640-01-235-1378	PUMP, CIRCULATING, 120/230 Volt, 50/60 Hz, AC

The circulating pump is intended to be used with the following water bath for thawing frozen blood units:

NSN	Item Identification
6640-01-237-0564	WATER BATH, ELECTRIC, 120/230 Volts, 50/60 Hz, AC

MILITARY INTERESTS:

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review Activities:
Army - MD
Navy - MS
Air Force - 03

User Activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:

DOD-MB

Agent:

DLA-DM

Project No. 6640-1532

Location: Disk No. 14
MIL/SPEC/PUMP/S31

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-42063(DM)	PUMP, CIRCULATING, 120/230 Volt, 50/60 Hz AC

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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